



# Certified Penetration Testing Professional

## Snort Rules Cheat Sheet

### Snort

Source: <https://www.snort.org>

Snort is an open-source network intrusion detection system, that can perform real-time traffic analysis and packet logging on IP networks. Snort's rule engine enables custom rules to meet the needs of the network. Snort rules help in differentiating between normal internet activities and malicious activities.

### Structure of a Snort Rule

|             |              |
|-------------|--------------|
| Rule Header | Rule options |
|-------------|--------------|

Basic structure of Snort rules

|        |          |         |      |           |         |      |
|--------|----------|---------|------|-----------|---------|------|
| Action | Protocol | Address | Port | Direction | Address | Port |
|--------|----------|---------|------|-----------|---------|------|

Structure of Snort rule header

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## 1. Rule Header

| Rule Actions                                                                            |                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Action                                                                                  | Description                                                                                                                                                              |
| Pass                                                                                    | Tells Snort to ignore the packet                                                                                                                                         |
| Log                                                                                     | To log a packet                                                                                                                                                          |
| Alert                                                                                   | To create an alert and then to activate another rule for checking more conditions                                                                                        |
| block                                                                                   | Block the current packet and all the subsequent packets in this flow                                                                                                     |
| drop                                                                                    | Drop the current packet                                                                                                                                                  |
| react                                                                                   | Send response to client and terminate session                                                                                                                            |
| reject                                                                                  | Terminate session with TCP reset or ICMP unreachable                                                                                                                     |
| rewrite                                                                                 | Enables overwriting packet contents based on a 'replace' option in the rules                                                                                             |
| Protocols                                                                               |                                                                                                                                                                          |
| Rule                                                                                    | Description                                                                                                                                                              |
| alert icmp any any -> any any (msg: "Ping with TTL=100"; \ ttl: 100;)                   | Shows on which type of packet the rule will be applied                                                                                                                   |
| The Direction Operator                                                                  |                                                                                                                                                                          |
| log !<Target Ip1>/24 any <> <Target Ip2>/24 23                                          | <> symbol shows that the rule will be applied to packets traveling on either direction.                                                                                  |
| alert tcp any any -> <Target Ip>/24 any (flags: A; \ ack: 0; msg: "TCP ping detected";) | -> shows that address and port numbers on the left-hand side of the direction field are the sources of the packet while those on the right-hand side are the destination |
| Port Numbers                                                                            |                                                                                                                                                                          |
| Port ranges                                                                             |                                                                                                                                                                          |
| alert udp any 1024:2048 -> any any (msg: "UDP ports";)                                  | Use a range of ports instead of only one port in the port field                                                                                                          |
| Negation symbol                                                                         |                                                                                                                                                                          |
| log udp any !53 -> any any log udp                                                      | Using the negation symbol (!) with port numbers to exclude a port or a range of ports                                                                                    |

| <b>Addresses</b>                                                                                                                |                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <code>alert tcp any any -&gt; &lt;Target Ip&gt;/32 80 (msg: "TTL=100"; \ ttl: 100;)</code>                                      | Two address parts in a Snort rule are used to check the source and the destination of the packet |
| <b>Address exclusion</b>                                                                                                        |                                                                                                  |
| <code>alert icmp ![&lt;Ip range&gt;] any -&gt; any any \ (msg: "Ping with TTL=100"; ttl: 100;)</code>                           | Snort provides a mechanism to exclude addresses using the negation symbol an exclamation point.  |
| <b>Address lists</b>                                                                                                            |                                                                                                  |
| <code>alert icmp ![ &lt;Ip range&gt; /24, &lt;Ip range&gt;/24] any -&gt; any \ any (msg: "Ping with TTL=100"; ttl: 100;)</code> | Specifies a list of addresses in a Snort rule                                                    |

## 2. Rule Options

| <b>Option</b>                                                                                                                     | <b>Description</b>                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>ACK keyword</b>                                                                                                                |                                                                                                     |
| <code>alert tcp any any -&gt; &lt;IP range&gt; any (flags: A; \ ack: 0; msg: "TCP ping detected");)</code>                        | Alerts when you receive a TCP packet with the ACK flag set and it contains a value of 0             |
| <b>classtype Keyword</b>                                                                                                          |                                                                                                     |
| <code>alert udp any any -&gt; &lt;IP range&gt; 6838 (msg:"DoS"; \ content: "server"; classtype:DoS;)</code>                       | Escalate high-risk alerts such as DoS attempts                                                      |
| <b>content Keyword</b>                                                                                                            | Find the signatures in the packet                                                                   |
| <code>alert tcp &lt;IP range&gt; any -&gt; &lt;IP range&gt; any \ (content: "GET"; msg: "GET matched");)</code>                   | Find the signatures in the packet                                                                   |
| <b>offset Keyword</b>                                                                                                             |                                                                                                     |
| <code>alert tcp &lt;IP range&gt; any -&gt; any any \ (content: "HTTP"; offset: &lt;no of bytes&gt;; msg: "HTTP matched");)</code> | Starts searching for the word "HTTP" after the specified number of bytes from the start of the data |

| depth Keyword                                                                                                                                             |                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <pre>alert tcp &lt;IP range&gt;any -&gt; any any (content: \ "HTTP"; offset: &lt;no of bytes&gt;; depth: &lt;no of chars&gt;; msg: "HTTP matched";)</pre> | Search the word "HTTP" between the specified range of the data part of the TCP packet  |
| content-list Keyword                                                                                                                                      |                                                                                        |
| <pre>alert ip any any -&gt; &lt;IP range&gt; any (content-list: \ "yourtext"; msg: "Text matched";)</pre>                                                 | Search the given strings in the data portion of all packets matching the rule criteria |
| dsize Keyword                                                                                                                                             |                                                                                        |
| <pre>alert ip any any -&gt; &lt;IP range&gt; any (dsize: &gt; 6000; \ msg: "Large size IP packet detected";)</pre>                                        | Alerts if the data size of an IP packet is larger than 6000 bytes                      |
| flags Keyword                                                                                                                                             |                                                                                        |
| <pre>alert tcp any any -&gt; &lt;IP range&gt; any (flags: SF; \ msg: "SYN- FIN packet detected";)</pre>                                                   | Alerts any scan attempt using SYN-FIN TCP packets                                      |
| fragbits Keyword                                                                                                                                          |                                                                                        |
| <pre>alert icmp any any -&gt; &lt;IP range&gt; any (fragbits: D; \ msg: "Don't Fragment bit set";)</pre>                                                  | Alerts if the DF bit is set in an ICMP packet                                          |
| icmp_id Keyword                                                                                                                                           |                                                                                        |
| <pre>alert icmp any any -&gt; any any icmp_id: 100; \ msg: "ICMP ID=100";)</pre>                                                                          | Verifies if the ICMP ID field in the ICMP header is equal to 100                       |
| icmp_seq Keyword                                                                                                                                          |                                                                                        |
| <pre>alert icmp any any -&gt; any any (icmp_seq: 100; \ msg: "ICMP Sequence=100";)</pre>                                                                  | Alerts if it finds a sequence number of 100                                            |
| itype Keyword                                                                                                                                             |                                                                                        |
| <pre>alert icmp any any -&gt; any any itype: 4; \ msg: "ICMP Source Quench Message received";)</pre>                                                      | Alerts if it detects a source quench message                                           |

| icode Keyword                                                                                                                                                                                                                                    |                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>alert icmp any any -&gt; any any itype: 5; \ icode: 1; msg: "ICMP ID=100";)</pre>                                                                                                                                                           | Alerts if host redirect ICMP packets are received                                                                                                     |
| ipopts Keyword                                                                                                                                                                                                                                   |                                                                                                                                                       |
| <pre>alert ip any any -&gt; any any (ipopts: lsrr; \ msg: "Loose source routing attempt");)</pre>                                                                                                                                                | Detects and alerts Loose Source Routing attempts                                                                                                      |
| ip_proto Keyword                                                                                                                                                                                                                                 |                                                                                                                                                       |
| <pre>alert ip any any -&gt; any any (ip_proto: ipip; \ msg: "IP-IP tunneling detected");)</pre>                                                                                                                                                  | Alerts if IPIP protocol is used by data packets                                                                                                       |
| logto Keyword                                                                                                                                                                                                                                    |                                                                                                                                                       |
| <pre>alert icmp any any -&gt; any any (logto:&lt;filename&gt;; ttl: 100;)</pre>                                                                                                                                                                  | Records all ICMP packets having TTL value equal to 100 to the specified file                                                                          |
| priority Keyword                                                                                                                                                                                                                                 |                                                                                                                                                       |
| <pre>alert ip any any -&gt; any any (ipopts: lsrr; \ msg: "Loose source routing attempt"; priority: 10;)</pre>                                                                                                                                   | Assigns a priority to a rule                                                                                                                          |
| react Keyword                                                                                                                                                                                                                                    |                                                                                                                                                       |
| <pre>alert tcp &lt;IP range&gt; any -&gt; any 80 (msg: "Outgoing \ HTTP connection"; react: block;)</pre>                                                                                                                                        | Terminate a session to block some sites or services                                                                                                   |
| reference Keyword                                                                                                                                                                                                                                |                                                                                                                                                       |
| <pre>alert udp \$EXTERNAL_NET any -&gt; \$HOME_NET 1900 \ (msg:"MISC UPNP malformed advertisement"; \ content:"NOTIFY * "; nocase; classtype:misc-attack; \ reference:cve,CAN-2001-0876; reference:cve, \ CAN-2001-0877; sid:1384; rev:2;)</pre> | Alert shows a reference to find more information about this alert. Adds a reference to information present on other systems available on the Internet |

| resp Keyword                                                                                                                          |                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <code>alert tcp any any -&gt; &lt;IP range&gt; 8080 (resp: rst_snd;)</code>                                                           | Sends a TCP Reset packet to the sender if an attempt is made to connect to TCP port 8080 on the local network |
| rpc Keyword                                                                                                                           |                                                                                                               |
| <code>alert ip any any -&gt; 192.168.1.0/24 any (rpc: 10000,* ,3; \ msg: "RPC request to local network";)</code>                      | Detects and alerts RPC based requests                                                                         |
| sameip Keyword                                                                                                                        |                                                                                                               |
| <code>alert ip any any -&gt; &lt;IP range&gt; any (msg: "Same IP"; \ sameip;)</code>                                                  | Alerts if source and destination IP addresses are same                                                        |
| session Keyword                                                                                                                       |                                                                                                               |
| <code>log tcp any any -&gt; &lt;IP range&gt; 110 (session: printable;)</code>                                                         | Rule logs all printable data from POP3 sessions                                                               |
| tag Keyword                                                                                                                           |                                                                                                               |
| <code>alert tcp &lt;IP range&gt; 23 -&gt; any \ (content: "facebook"; msg: "Detected facebook"; \ tag: session, 100, packets;)</code> | Records 100 packets on the session after it is detected                                                       |

### 3. Command Line Options

**Syntax:** snort [options]

| Options                                 | Description                                                                             |
|-----------------------------------------|-----------------------------------------------------------------------------------------|
| <code>-A alert-mode</code>              | Generates an alert using one of the specified alert-modes: fast, full, none, and unlock |
| <code>-b</code>                         | Logs packets in tcpdump format (i.e., libpcap)                                          |
| <code>-B address-conversion-mask</code> | Scrambles the networks specified in the -h (or HOME_NET) setting                        |
| <code>-c config-file</code>             | Allows you to specify which configuration file you want to use                          |
| <code>-C</code>                         | Prints the character data found in the packet payload                                   |

|                             |                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------|
| <b>-d</b>                   | Displays the application layer data when in verbose or packet logging mode         |
| <b>-D</b>                   | Turns Snort in daemon mode                                                         |
| <b>-e</b>                   | Displays or logs the link-layer packet headers                                     |
| <b>-F bpf-file</b>          | Reads Berkeley Packet Filters (BPF) from a bpf file                                |
| <b>-g group</b>             | Changes the default group ID or GID under which Snort runs after initialization    |
| <b>-h home-net</b>          | Sets the "home network" to a specific address in CIDR format                       |
| <b>-i interface</b>         | Specifies which interface Snort should listen on                                   |
| <b>-I</b>                   | In alerts, displays the interface on which each packet arrived                     |
| <b>-k checksum-mode</b>     | Controls which packet checksums Snort computes and verifies                        |
| <b>-l logging-directory</b> | Specifies the logging directory                                                    |
| <b>-L binary-log-file</b>   | Sets the filename of the binary log file                                           |
| <b>-m umask</b>             | Sets the file mode creation mask to the designated umask variable                  |
| <b>-n packet-count</b>      | Processes the given number of packets and then exits                               |
| <b>-N</b>                   | Turns off packet logging                                                           |
| <b>-o</b>                   | Changes the order in which the rules are applied to packets                        |
| <b>-O</b>                   | Replaces the IP addresses printed to the screen or log file with "xxx.xxx.xxx.xxx" |
| <b>-p</b>                   | Turns off promiscuous mode sniffing                                                |
| <b>-P snap-length</b>       | Sets the maximum packet capture length to a certain size                           |
| <b>-q</b>                   | Tells Snort to run quietly. Does not display banner and initialization information |
| <b>-r tcpdump-file</b>      | Processes a tcpdump-formatted file                                                 |
| <b>-s</b>                   | Sends alert messages to a syslog server                                            |

|                          |                                                                                |
|--------------------------|--------------------------------------------------------------------------------|
| <b>-S variable=value</b> | Sets the variable name <variable> to the value <value>                         |
| <b>-t chroot</b>         | Changes Snort's root directory to chroot after initialization                  |
| <b>-T</b>                | Starts Snort in self-test mode.                                                |
| <b>-u user</b>           | Changes the default user ID or UID under which Snort runs after initialization |
| <b>-U</b>                | Forces the timestamp in all logs to be in UTC format                           |
| <b>-v</b>                | Prints all packets to the console                                              |
| <b>-V</b>                | Displays the Snort version number and then exits                               |
| <b>-X</b>                | Displays raw packet data starting at the link layer                            |
| <b>-y</b>                | Includes the year in all alerts and log files                                  |
| <b>-z</b>                | Enables the stream4 preprocessor                                               |
| <b>-?</b>                | Lists all switches and options and then exits                                  |

## 4. Sniffer Mode

| Sniff packets and send to standard output as a dump file |                                                |
|----------------------------------------------------------|------------------------------------------------|
| Options                                                  | Description                                    |
| <b>-v (verbose)</b>                                      | Display output on the screen                   |
| <b>-e</b>                                                | Display link-layer headers                     |
| <b>-d</b>                                                | Display packet data payload                    |
| <b>-x</b>                                                | Display full packet with headers in HEX format |

## 5. Packet Logger Mode

| Input output to a log file |                                                   |
|----------------------------|---------------------------------------------------|
| Options                    | Description                                       |
| <b>-r</b>                  | Use to read back the log file content using snort |
| <b>-l (directory name)</b> | Log to a directory as a tcpdump file format       |
| <b>-k (ASCII)</b>          | Display output as ASCII format                    |



## 6. NIDS Mode

| Use the specified file as config file and apply rules to process captured packets |                                                    |
|-----------------------------------------------------------------------------------|----------------------------------------------------|
| Options                                                                           | Description                                        |
| -c                                                                                | Define configuration file path                     |
| -T                                                                                | Use to test the configuration file including rules |